Name: _____

Physical and chemical Changes

1. The physical properties of a substance are its observable characteristics. We can often identify substances by looking at them or feeling them. Unscramble the words below to find examples of physical properties.

CROLOU	_____	
PASHE	_____	
SARDHENS	_____	
SATET	_____	
DOUOR	_____	
TTSAE FO REMATT	_____	
LOBGINI	_____	Point
ZERFENGI	_____	Point

2. A physical change is a change in the physical characteristics of a substance. Physical changes happen without any change in the chemical makeup of the substance.

   Give 4 examples of physical changes.

   1.
   2.
   3.
   4.

3. Chemical changes are changes in matter that produce a new substance. They occur when substances react to form new substances with different properties to the original ones.

   List some signs that a chemical change may have taken place:

   1.
   2.
   3.
   4.

   Give 4 examples of chemical changes.

   1.
   2.
   3.

4.

4. Which of the following are physical changes and which are chemical changes?

a) Cutting your hair. _____
b) Copper turning green (tarnishing). _____
c) Peeling an orange. _____
d) Folding paper. _____
e) Burning paper. _____
f) Bursting a milk carton. _____
g) Firing a flare. _____
h) Melting ice. _____
i) Boiling water. _____
j) Rusting iron. _____
k) Cooking an egg. _____
l) Milk going off. _____
m) Mixing cordial and water. _____
n) Magnetising a pin. _____
o) A cut apple going brown. _____
p) Crushing a sugar cube. _____
q) Baking soda reacts with vinegar and forms a gas. _____
r) Mixing two clear liquids and a yellow colour forms. _____
s) A piece of metal is bent in half. _____
t) Diamonds are used to scratch glass. _____
u) Mixing butter, sugar and flour together. _____
v) Baking the mixture above into cookies. _____

5. When camping in the mountains, you boil a pot of water and cook pasta on a campfire. What physical and chemical changes take place in this process? Give reasons for your answers.